

Dr. Momo Klara Ellwarth

✉ momo.ellwarth@gmail.com ☎ +1 520 491 9117

🌐 [linkedin.com/in/momoellwarth](https://www.linkedin.com/in/momoellwarth)

Professional Experience

01/2025 – today

Flagstaff, AZ, USA

Postdoctoral Research Associate

LOWELL OBSERVATORY

- Observational solar physics using EXPRES/LOST high-resolution spectra to study the Sun as a star
- Development and validation of new spectroscopic activity diagnostics from EXPRES/LOST solar spectra to connect magnetic variability, line-profile changes, and radial-velocity signals

03/2024 – 12/2024

Göttingen, DE

Postdoctoral Researcher

INSTITUT FÜR ASTROPHYSIK UND GEOPHYSIK (IAG), UNIVERSITÄT GÖTTINGEN

- Analysis of line bisectors and formation-height diagnostics in high-resolution spatially resolved solar FTS spectra.

06/2019 – 03/2024

Göttingen, DE

Doctoral Researcher/Research Assistant

INSTITUT FÜR ASTROPHYSIK UND GEOPHYSIK (IAG), UNIVERSITÄT GÖTTINGEN

- Development of a calibration and processing workflow for spatially resolved solar FTS observations, resulting in a high-resolution center-to-limb quiet Sun atlas
- Line-by-line spectral analysis to investigate center-to-limb variations of convective blueshifts, line-depth dependence, and chromatic effects

09/2013 – 07/2017

Freiburg i. Br., DE

Undergraduate/Graduate Research Assistant

INSTITUTE OF SOLAR PHYSICS - KIS

- Developed test routines for the Image Stabilization System of Solar Orbiter/PHI, including laboratory work on the optical setup

Education

06/2019 – 03/2024

Göttingen, DE

PhD (Dr. rer. nat.) in Physics (*Grade: magna cum laude*)

GEORG AUGUST UNIVERSITY GÖTTINGEN

Advisor: Prof. Dr. Ansgar Reiners

- Dissertation: *High-Precision Spectroscopy of the Solar Surface*

09/2015 – 09/2018

Freiburg i. Br., DE

M. Sc. Applied Physics (*Grade: 1.9; Masterthesis: 1.7*)

ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

Advisor: Prof. Dr. Wolfgang Schmidt

09/2011 – 07/2015

Freiburg i. Br., DE

B. Sc. Physics (*Grade: 2.7; Bachelorthesis: 1.7*)

ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

09/2009 – 06/2011 Bad Säckingen, DE	Abitur (German university entrance qualification), focus on Physics TECHNISCHE OBERSCHULE
09/2006 – 07/2009 Steinen, DE	Vocational Training as Food Retail Sales Assistant METZGEREI HUG

Publications

First Author

- 11/2025 The He I D3 Line as a Proxy for Magnetic Activity Using EXPRES Solar Observations**
M. ELLWARTH, J. LLAMA, L. L. ZHAO, J. M. BREWER, A. SZYMKOWIAK, R. M. ROETTENBACHER, J. T. SIKORA, A. S. POLANSKI, S. H. SAAR;
 THE ASTROPHYSICAL JOURNAL, VOLUME 994, ID.104, 8 PP.; 2025APJ...994..104E
- 12/2023 Convective characteristics of Fe I lines across the solar disc**
M. ELLWARTH, B. EHMANN, S. SCHÄFER, A. REINERS;
 ASTRONOMY & ASTROPHYSICS, VOLUME 680, ID.A62, 10 PP.; 2023A&A...680A..62E
- 05/2023 The IAG spectral atlas of the spatially resolved Sun: Centre-to-limb observations**
M. ELLWARTH, S. SCHÄFER, A. REINERS, M. ZECHMEISTER;
 ASTRONOMY & ASTROPHYSICS, VOLUME 673, ID.A19, 12 PP.; 2023A&A...673A..19E
- 09/2021 Newly formed downflow lanes in exploding granules in the solar photosphere**
M. ELLWARTH, C. E. FISCHER, N. VITAS, S. SCHMIZ, W. SCHMIDT;
 ASTRONOMY & ASTROPHYSICS, VOLUME 653, ID.A96, 13 PP.; 2021A&A...653A..96E

Co-Author

- 03/2026 Solar carbon abundance from 3D non-LTE modelling of the diagnostic lines of the CH molecule**
 R. HOPPE, M. BERGEMANN, P. EITNER, **M. ELLWARTH**, Å. NORDLUND, J. LEENAARTS, B. PLEZ, A. SERENELLI;
 MONTHLY NOTICES OF THE ROYAL ASTRONOMICAL SOCIETY, VOLUME 546, ISSUE 3, ID.STAF2085, 22 PP.; 2026MNRAS.546F2085H
- 10/2025 Solar flux atlases: The new HARPS-N quiet Sun benchmark and continuum normalisation of the Ca II H & K lines**
 F. HANASSI-SAVARI, A. G. M. PIETROW, M. K. DRUETT, M. CRETIGNIER, **M. ELLWARTH**;
 ASTRONOMY & ASTROPHYSICS, VOLUME 702, ID.A97, 22 PP. ; 2025A&A...702A..97H
- 01/2025 Precise and efficient modeling of stellar-activity-affected solar spectra using SOAP-GPU**
 Y. ZHAO, X. DUMUSQUE, M. CRETIGNIER, K. AL MOULLA, **M. ELLWARTH**, A. REINERS, A. SOZZETTI;
 ASTRONOMY & ASTROPHYSICS, VOLUME 693, ID.A262, 13 PP. ; 2025A&A...693A.262Z
- 05/2023 Solar center-to-limb variation in Rossiter-McLaughlin and exoplanet transmission spectroscopy**
 A. REINERS, F. YAN, **M. ELLWARTH**, H.-G. LUDWIG, L. NORTMANN;
 ASTRONOMY & ASTROPHYSICS, VOLUME 673, ID.A71, 11 PP.; 2023A&A...673A..71R

- 12/2021** **Solar oxygen abundance**
M. BERGEMANN, R. HOPPE, ET AL., INCLUDING **M. ELLWARTH**;
MONTHLY NOTICES OF THE ROYAL ASTRONOMICAL SOCIETY, VOLUME 508, ISSUE 2,
PP.2236-2253; 2021MNRAS.508.2236B
- 12/2020** **Observing the integrated and spatially resolved Sun with ultra-high spectral resolution**
S. SCHÄFER, K. ROYEN, A. HUSTER ZAPKE, **M. ELLWARTH**, A. REINERS;
PROCEEDINGS OF THE SPIE, VOLUME 11447, ID. 11447A9 22 PP. (2020); 2020SPIE11447E..A9S
- 08/2016** **End-to-end simulations of the visible tunable filter for the Daniel K. Inouye Solar Telescope**
W. SCHMIDT, M. SCHUBERT, **M. ELLWARTH**, ET AL.;
PROCEEDINGS OF THE SPIE, VOLUME 9908, ID. 99084N 11 PP. (2016).; 2016SPIE.9908E..4NS

Selected Talks, Presentations & Outreach

*Invited or solicited talk

*International

- 01/2026** **245th Meeting of the American Astronomical Society (AAS)⁺**
Phoenix, AZ, USA LINE-BY-LINE DIAGNOSTICS OF SOLAR ACTIVITY
- 12/2025** **KIS Seminar***
Institute of Solar Physics (KIS)
Freiburg i. Br., DE SOLAR ACTIVITY DEPENDENCE OF SPECTRAL LINES: A LINE-BY-LINE ANALYSIS
- 07/2025** **Lowell Observatory Pluto Circle Event**
Lowell Observatory
Flagstaff, AZ, USA ILLUMINATING THE SUN: A STELLAR KEY TO OTHER EARTHS
PUBLIC OUTREACH TALK FOR LOWELL OBSERVATORY MEMBERS
- 07/2025** **EPRV 6⁺**
Porto, PT SOLAR ACTIVITY DEPENDENCE OF SPECTRAL LINES IN EXPRES DATA
- 03/2025** **Flagstaff Astronomy Symposium 2025**
Lowell Observatory
Flagstaff, AZ, USA ILLUMINATING THE SUN: INSIGHTS INTO CONVECTIVE MOTIONS
- 10/2024** **PoET Workshop # 2^{*+}**
Porto, PT VARYING CONVECTIVE MOTIONS IN HIGH-RESOLUTION IAG SOLAR OBSERVATIONS
SOLICITED CONSORTIUM TALK
- 03/2024** **SPP 1992 Final Colloquium***
Berlin, DE UNDERSTANDING RADIAL VELOCITY JITTER: BENCHMARK OBSERVATIONS OF THE SUN
SOLICITED CONSORTIUM TALK
- 05/2024** **Thüringer Landessternwarte Colloquium***
Thüringer Landessternwarte
Tautenburg, DE INVESTIGATING CONVECTIVE MOTIONS OF THE SPATIALLY RESOLVED QUIET SUN
- 02/2024** **Max Planck Institute for Solar System Research (MPS); public lecture series***
MPS
Göttingen, DE *Was ist da draußen?*
PRÄZISIONSMESSUNGEN MIT STERNENLICHT: VON EXOPLANETEN, PHYSIKALISCHEN KONSTANTEN UND DER EXPANSION DES UNIVERSUMS

03/2024 Penn State University Penn State, PA, USA	CEHW Seminar* (online) CENTRE-TO-LIMB VARIATIONS OVER THE SOLAR DISC: A SPECTRAL ATLAS OF THE SPATIALLY RESOLVED SUN
03/2023 Santa Barbara, CA, USA	EPRV 5⁺ DECOMPOSING STELLAR VARIABILITY: A SPECTRAL ATLAS OF THE SPATIALLY RESOLVED SUN
09/2022 Berlin, DE	PFE-SPP 1992 Joint Meeting HIGH RESOLUTION SPECTROSCOPY OF THE SPATIALLY RESOLVED SUN

Grants & Funding

2021 – 2022 Geneva, CH	SPP 1992 Visitor Program 2021/22 AWARDED 5,400€ TO SUPPORT A THREE-MONTH RESEARCH VISIT AT OBSERVATOIRE DE GENÈVE
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Scientific Collaborations & Consortia

2025 – today	EXPRES/LOST Solar Collaboration POSTDOCTORAL RESEARCHER CONTRIBUTING TO SOLAR ACTIVITY DIAGNOSTICS USING EXPRES SUN-AS-A-STAR OBSERVATIONS
2024 – today	Paranal solar ESPRESSO Telescope (PoET) MEMBER OF THE SCIENCE TEAM

Research Visits

11/2021 - 02/2022 Versoix, CH	Visiting Researcher OBSERVATOIRE DE GENÈVE <i>Host: Prof. Dr. Xavier Dumusque</i>
09/2019 Boulder, CO, USA	Visiting Researcher HIGH ALTITUDE OBSERVATORY <i>Host: Dr. Johannes Löhner-Böttcher</i>

Teaching Experience

STUDENTS - Research supervision and mentorship

2024 – 2025 University Göttingen	Laura Losch BACHELOR THESIS - MODELING OF GRANULAR FLOW AND ITS DOPPLER SHIFT ACROSS THE SOLAR DISC
2022 – 2023 University Göttingen	Birte Ehmann MASTER THESIS - CENTER-TO-LIMB VARIATIONS OF CONVECTIVE BLUESHIFTS IN THE SOLAR ATMOSPHERE
2020 – 2021 University Göttingen	Christoph Fischer BACHELOR THESIS - HIGH-RESOLUTION SPECTROSCOPIC MEASUREMENTS OF CONVECTIVE BLUESHIFTS ON THE SOLAR SURFACE

COURSES

2021 – 2022 **Experimental Physics II: Electromagnetism**
University Göttingen TEACHING ASSISTANT

2020 – 2021 **Experimental Physics I: Mechanics**
University Göttingen TEACHING ASSISTANT

MISCELLANEOUS

2024 **High-School Internship Project**
University of Göttingen DEVELOPED AND SUPERVISED A PROGRAMMING-BASED SOLAR PHYSICS EXERCISE ON MEASURING THE SOLAR ROTATION RATE

2012 – 2019 **Private Tutor in Mathematics and Physics**
INDIVIDUAL INSTRUCTION FOR SECONDARY-SCHOOL AND UNIVERSITY-LEVEL STUDENTS

Academic Service & Student Representation

Ad hoc reviewer for *Astronomy & Astrophysics (A&A)*

since 2025 **Member**
USA AMERICAN ASTRONOMICAL SOCIETY (AAS)

since 2020 **Member**
Germany GERMAN PHYSICAL SOCIETY (DPG)

10/2012 - 03/2016 **Active Member of the Departmental Student Association**
University of Freiburg STUDENT REPRESENTATION, COMMUNICATION WITH FACULTY LEADERSHIP, AND CONTRIBUTION TO DEPARTMENTAL REGULATIONS AND INTERNAL PROCESSES

10/2012 - 10/2014 **Elected Member of the University Student Council**
University of Freiburg UNIVERSITY-WIDE STUDENT REPRESENTATION, BUDGET DECISIONS, AND ORGANIZATION OF STUDENT EVENTS

10/2013 - 07/2014 **Deputy Member of the Study Commission**
University of Freiburg REPRESENTATION IN TEACHING, CURRICULUM, AND EXAMINATION MATTERS

Technical Skills

- **Programming & Data Analysis:** Python, IDL, C/C++
- **Scientific Computing:** Jupyter, Git, GitHub, Linux/Unix shell, $\text{\LaTeX}$, Overleaf
- **Languages:** German (nativ), English (fluent), Spanish (basic)